Academic English L2 - Using graphs and charts

Choosing the right noun

Noun	Remember	Simple explanation
Amount	How much?	For things we do not usually count one by one: time, money, food.
Level	How high?	For something that can be low or high: stress, satisfaction, noise.
Number	How many?	For separate things we can count: students, books, applications.
Quantity	How much material?	Often for something measured or used: food, paper, equipment.
Rate	Per what?	A result measured per person, per meal, per year, or for a defined group.
Share	What part?	The part belonging to one group out of the whole, often as a percentage.
Size	How big?	How large a class, group, sample, or university is.
Volume	How much activity?	The total activity, such as loans or applications, during a period.

Examples from research

These sentences are adapted from the four studies listed at the end. The letters identify the source of each example.

Amount

1. Many chemistry students had less study time than their teachers expected: 37.3% had only two to four hours available each week. The **amount of time** they had was therefore an important part of the study. [C]
2. For some students, paid work reduced the **amount of time** available for studying. [C]
3. In the university canteens studied, the average **amount of food** thrown away was about 72 grams per student per meal. [G]

Level

1. The Zambian researchers asked students about their **level of satisfaction** with online learning. [M]
2. Students over 36 were more likely than younger students to report a higher **level of satisfaction** with online classes. [M]
3. Students in the chemistry study reported very different **levels of available study time**: some had fewer than two hours a week, while others had more than seven. [C]

Number

1. The **number of students** included in the chemistry study was 110, although 132 were enrolled in the course. [C]
2. Of the 737 students in the Zambian survey, the **number of women** was 380. [M]
3. The researchers examined a striking **number of library records**: 118,159 in total. [Z]

Quantity

1. To study food waste, the researchers recorded the **quantity of food** students left after a meal. [G]
2. Across 29 universities, the average **quantity of food wasted** came to about 72 grams per student per meal. [G]
3. By looking at the **quantity of staple foods and meat** thrown away, the researchers could compare patterns of waste between regions. [G]

Rate

1. Data were collected from 110 of the 132 enrolled chemistry students: a study participation **rate** of about 83%. [C]
2. In the Zambian study, 737 students took part after 750 questionnaires were distributed, giving a questionnaire participation **rate** of about 98%. [M]
3. The Chinese study estimated a food-waste **rate** of about 72 grams **per student per meal**. [G]

Share

1. The **share of surveyed students** who wanted blended learning to continue was 56.5%. [M]
2. An even larger **share** - 78.4% - thought group discussions worked better in a classroom. [M]
3. Only a small **share of the chemistry students**, 7.3%, studied for as many hours as their teachers recommended. [C]

Size

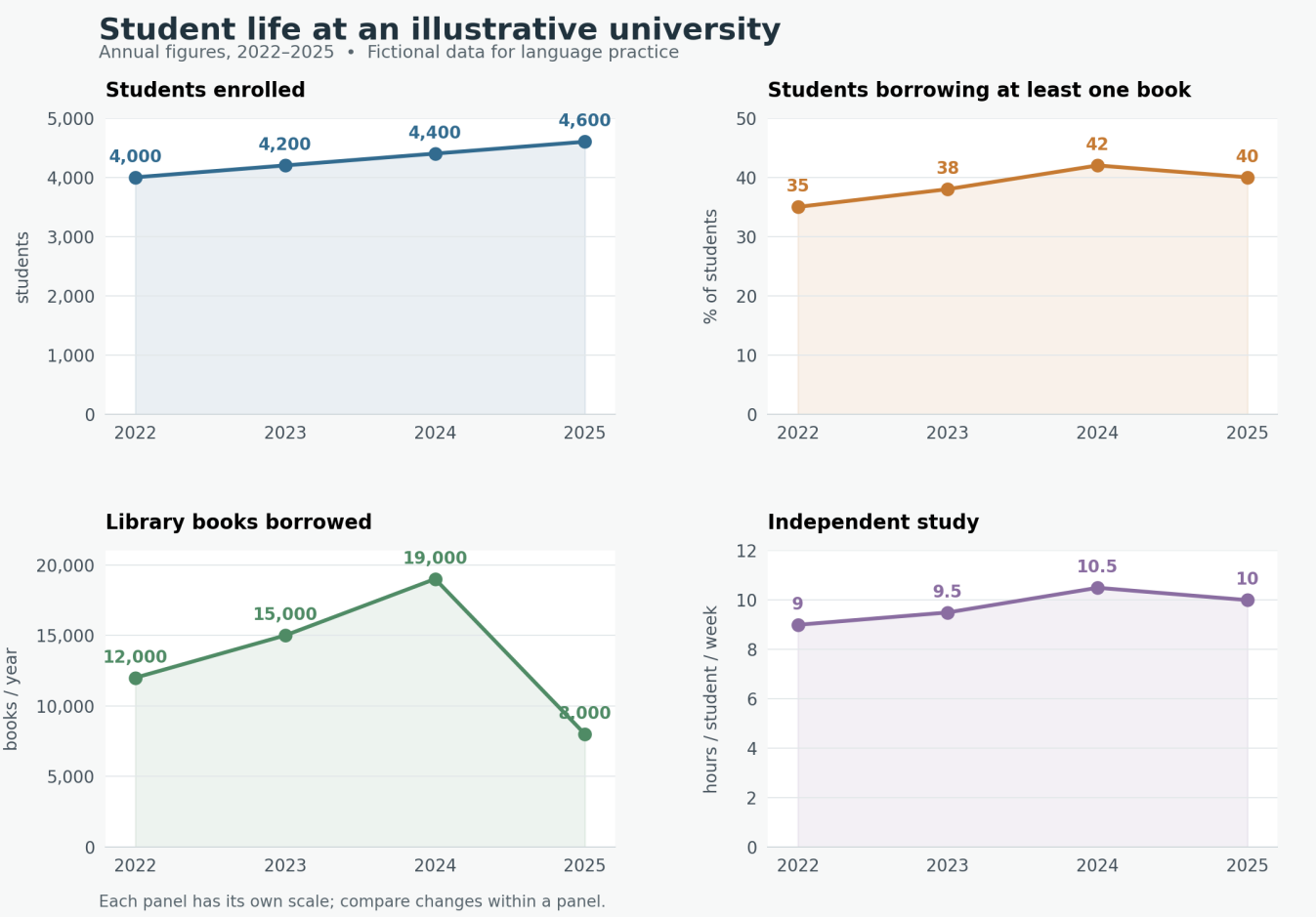
1. The researchers had planned for a **sample size** of at least 546 students; in the end, 737 took part. [M]
2. The **size of the library dataset** allowed the researchers to compare borrowing patterns across different university subjects. [Z]
3. With 9,192 participants from 29 universities, the Chinese food-waste study had a large **sample size**. [G]

Volume

1. Researchers had to analyse a large **volume of library records** - 118,159 borrowing records. [Z]
2. The **volume of borrowing** was particularly high among students in subjects including information technology and economics. [Z]
3. Male and female students generated almost the same **volume of book loans** in the library data. [Z]

Practice chart: changes over time

Fictional data for language practice.



Model description

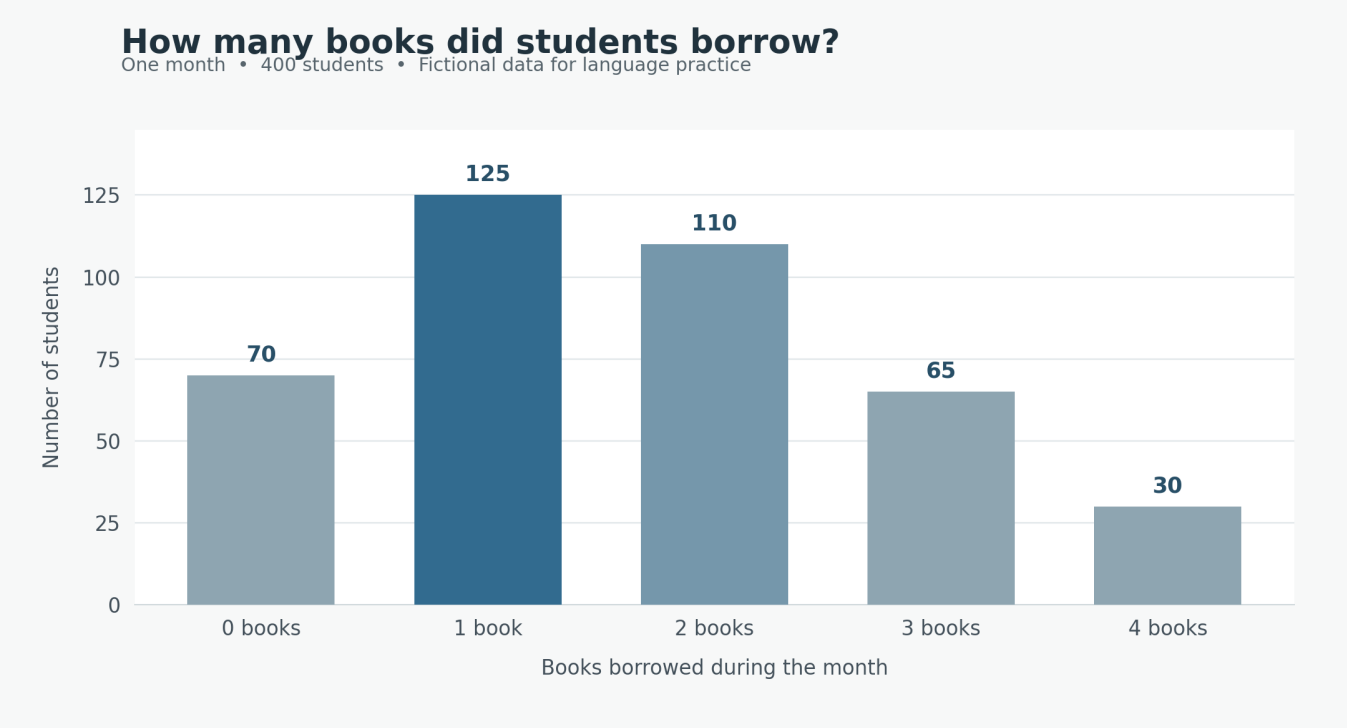
The **number** of students rose steadily from 4,000 in 2022 to 4,600 in 2025. In other words, the university grew in **size** by 15%. Its yearly growth **rate** stayed close to 5%.

The **share** of students who borrowed at least one library book rose from 35% to 42%, then fell slightly to 40% in 2025. However, the **volume** of book loans changed much more dramatically: after reaching 19,000 in 2024, it dropped to 8,000 the following year. That is a drastic decrease of about 58%. The **level** of borrowing was therefore far lower in 2025, even though almost the same share of students still borrowed a book.

Finally, the average **amount of time** spent studying independently increased from nine to 10.5 hours per student per week, before decreasing slightly to ten hours.

Practice chart: comparing groups

Fictional data for language practice.



Model description

The bar chart groups 400 students by the **number** of library books they borrowed during one month. The largest group, 125 students, took out one book; its **size** was slightly greater than that of the two-book group (110). Together, these groups account for a substantial **share** of students: 235 out of 400, or 58.75%. At the highest borrowing **level** shown, four books, there were just 30 students.

Across all groups, the total **volume** of borrowing was 660 book loans. This gives an average **rate** of 1.65 books per student per month. The chart measures the **quantity** of reading material borrowed, but it cannot tell us the **amount of time** students spent reading or how much they understood.

Sources for the example sentences

[C] Clement (2016), External and Internal Barriers to Studying Can Affect Student Success and Retention in a Diverse Classroom.

[M] Mudenda et al. (2023), Student's perspectives, satisfaction and experiences with online and classroom learning during the COVID-19 pandemic.

[Z] Zhang et al. (2024), Empirical study of college students' extracurricular reading preference by functional data analysis of the library book borrowing behavior.

[G] Gao et al. (2024), Lost Food and Associated Phosphorus Footprint: Evidence from China.